

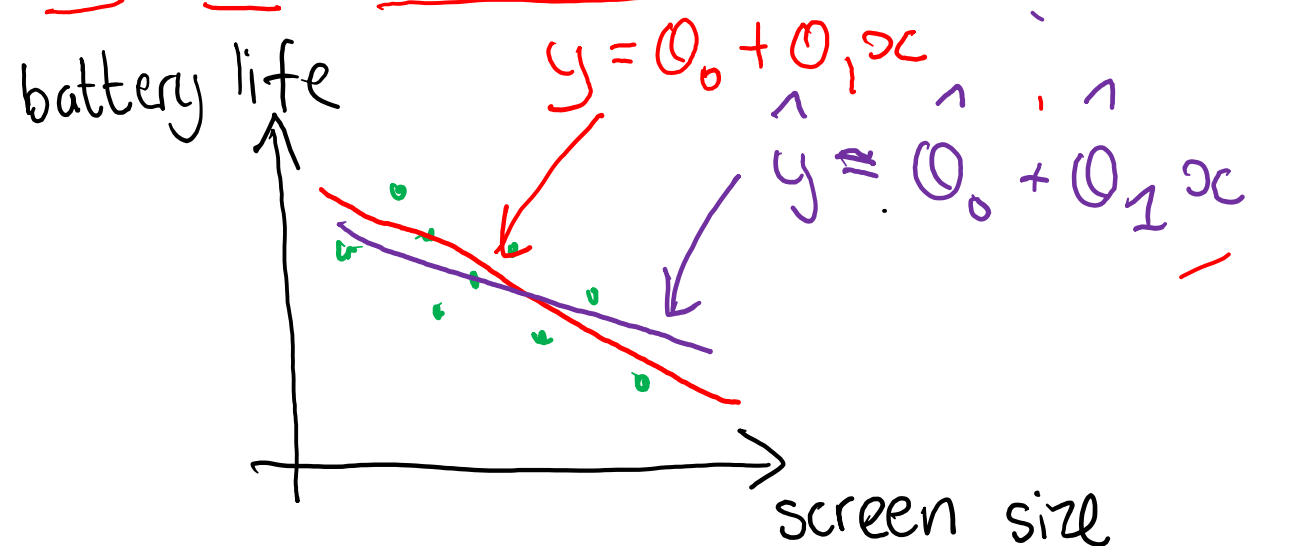
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Simple Linear Regression



Simple Linear Regression

- Assumes a linear model
- One output response value and a single input feature
- $h_{\theta}(x) = y \approx \theta_0 + \theta_1 x$ $y \approx b + mx$ $\hat{y} \approx \hat{\theta}_0 + \hat{\theta}_1 x$
- E.g. battery_life $\approx$ θ_0 + θ_1 $\times$ screen_size



Optimization

- Choose optimization metric (MSE)
- Choose optimization algorithm
 - Ordinary Least Squares (analytical solution)
 - Analytical solutions are not always easy to determine
 - Gradient Descent (numerical solution)
 - Intuitive approach

$$\begin{aligned}\underline{J(\theta_0, \theta_1)} &= \frac{1}{N} \sum_{i=1}^N (\underline{h_{\theta}(x_i)} - \underline{y_i})^2 \\ &= \frac{1}{N} \sum_{i=1}^N ((\underline{\theta_0} + \underline{\theta_1 x_i}) - y_i)^2 \\ &= \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2\end{aligned}$$



Next Up:

Gradient Descent

